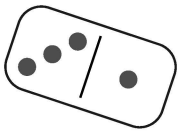
MathsJam Shout



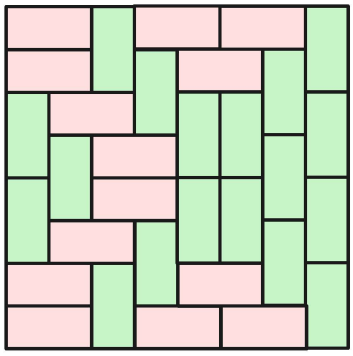


September 2025
Bletchley MathsJam, UK

This month's contribution from Bletchley MathsJam in Milton Keynes is **domino themed** - if you can get hold of a set of dominoes to use, bring them along (or else you can print the ones at **mathsj.am/printable-dominoes**).

Make Faultless Filling

Domino tiling fills a rectangle with 1x2 dominoes. This example 8x8 grid has a "fault line" at the right – where you could split the grid in two with a ruler without cutting any dominoes in half. But remove the four right-hand dominoes and no fault lines remain.



What's the smallest rectangle you can tile without any fault lines? In general, which sizes of rectangle can be tiled without faults?

Think Double Dominoes

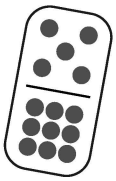
A set of dominoes consists of one of each unique pairing of dots – 0 to 6 dots on each side for a traditional "double six" set, or 0 to 9 for a "double nine" set.



How many dominoes are there in each set? How many dots are there in each? What about a "double n" set?

Puzzle Set Searching

In this classic "Dominosa" puzzle, the challenge is to identify how to divide the grid to create a full set of "double six" dominoes (everything from 0-0 to 6-6) – there's only one unique solution.



You might wish to make a list of which dominoes you've already found and draw a line between anywhere else that pair of numbers appears.

4	3	0	3	3	1	5	5
1	4	6	3	3	0	4	5
2	5	5	5	6	2	6	0
2	5	3	0	0	6	2	1
0	5	6	4	0	0	2	1
6	2	4	4	1	2	1	3
6	6	1	1	4	4	2	3

Play online: mathsj.am/dominosa

MathsJam Shout is a monthly sheet of ideas for activities to do at a MathsJam night. It's created using suggestions from a different MathsJam each month, and if you'd like to submit suggestions for a month in the future, email **katie@mathsjam.com** for details.

MathsJam is a monthly opportunity for like-minded self-confessed maths enthusiasts to get together in a pub and share stuff they like. Puzzles, games, problems, or just anything they think is cool or interesting. Monthly MathsJam nights happen in over 70 locations around the world, on the second-to-last Tuesday of each month. To find your nearest MathsJam, visit the website at **www.mathsjam.com**.

PRINTABLE DOMINOES

